A Lightweight Authentication Scheme with PE-Based Unclonable Label

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***Abstract*—With the maturity of printed electronics (PE) technology, the market of PE products grows rapidly. Meanwhile, as PE products usually carry plenty of sensitive information, the anti-counterfeiting has become an important security issue that cannot be ignored. Physical unclonable function (PUF) as an emerging security primitive that can be used to uniquely identify products, has received widespread attention. In this paper, we propose an optical unclonable label (OUL) compatible with the PE process and the pattern is flexible enough, released from the constraints of circuits, which only introduces slight area overhead suitable for the ultra-low-cost PE products. Then, a robust ID extraction algorithm is designed to build lightweight authentication system, which can effectively reduce storage cost while maintaining the uniqueness and reliability of OUL. In addition, a pseudo label recognition is added to resist forged labels, which utilizes the overall sample statistics to provide double insurance. To evaluate the effect of proposed method, 100 OULs are printed, receiving an overall identification certainty of 14.17σ and best figure of merit (FoM) of 0.29, which shows great attraction to the current anti-counterfeiting scheme.**

Keywords—PE; PUF; optical unclonable label (OUL); anti-counterfeiting; authentication

I. INTRODUCTION

Printed electronics (PE) as an emerging technology has received growing attention in recent years, which uses traditional printing techniques to produce electronic devices and circuits. Compared with subtractive manufacturing technology used in integrated circuits (ICs), PE is an additive manufacturing technology that enables the electronic devices to be fabricated with several advantages, such as large-scale, low-cost, customizable, mechanism flexible and environment friendly [1]. These advantages promote PE to be used in many frontier fields, such as the Internet of Thing (IoT) [2], smart sensors [3], smart packaging [4], and ingestible health diagnostic equipment [5]. It is reported that the global PE market is projected to reach $23 billion in 2026 from an estimated $9.9 billion in 2021, which indicates a compound annual growth rate of 18.3% from 2021 to 2026 [6]. The forecast of the PE market indicates the signification of PE technology for future electronic products, and means that the massive increase in PE products has become an inevitable trend.

With the widespread application of PE products, the counterfeiting of PE products can be predicted to intensify. Especially, in the fields of IoT and smart sensors, which involve massive sensitive information with high security requirements, effective countermeasures need to be explored. Therefore, how to prevent and identify the counterfeiting of PE products has become an urgent security issue. Generally, barcodes are used as an identifier to identify products [7], due to the rules of manufacturing barcodes are definitive, that cannot prevent them from being forged. Direct part marking, such as laser etching, has been widely used for industrial part identification [8]. However, for mass produced PE products, etching or embossing a mark on each product takes a large amount of time, which could become a bottleneck in the production line. Radio frequency identification (RFID) is another alternative option [9], while it contains specific chips that are susceptible to environmental influences. In addition, for ultra-low-cost PE products, the cost of attaching them to millions of products cannot be ignored as well.

Physical unclonable function (PUF) shed new light on the anti-counterfeiting, where the inevitable deviations in the manufacturing process can be used as a unique identifier of the product. Since these deviations are totally random and uncontrolled, they are difficult to be predicted or duplicated, making counterfeiting quite expensive and economically infeasible, which shows an ideal anti-counterfeiting property. Based on this, various forms of PUF schemes have been proposed and applied in authentication and key generation in the past two decades [10]-[12], among which optical PUFs have received extensive attention since their patterns contain rich entropy and high output complexity [12]. Optical PUFs use the emitted light to activate the implicit randomness in the object [10], then applies some image processing techniques to identify items, where the latter is a quite mature discipline. Moreover, since optical PUFs usually do not require to design specific circuits which shows great flexibility, the cost of optical PUFs can be effectively reduced, making them attractive in PE products.

In this work, we found that when aqueous nano-silver inks are printed on polyethylene terephthalate (PET) film, it exhibits complex patterns that are uncontrollable and unpredictable. This naturally formed pattern contains rich entropy, just like a natural "fingerprint". Inspired by this phenomenon, we design an optical unclonable label (OUL), which is compatible with PE technology and only introduces slight area overhead. Moreover, the pattern of OUL is flexible enough, enabling it release from the constrains of the circuit. Then, a lightweight authentication system is built, which contains ID extraction algorithm and pseudo label recognition, that the former can effectively reduce storage cost of OUL without compromising uniqueness and reliability, and the latter can provide double insurance against forged labels. To evaluate the effect of the proposed method, 100 OULs are printed, receiving an overall identification certainty of 14.17σ and best figure of merit (FoM) of 0.29, which shows great attraction to the current anti-counterfeiting scheme.

The rest of this paper is organized as follows. Section II reviews some existing work on PE-based PUFs. Section III introduces the details of proposed OUL and the overview of authentication system. Results and discussion are analyzed in Section IV. Finally, this paper is concluded in Section V.

II. RELATED WORKS

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A. Printed Electronics

PE technology combines electronic manufacturing and graphic printing, enabling the products printed on-demand, mechanism flexible, low-cost and environment friendly [1]. To realize high-quality electronic products which are soft, wearable and economic, suitable printable electronic solutions and substrates must be selected. Recently, various particular catalyst solutions and substrates have been researched and developed. Sun et al. [13] reported that dichloroethane and ethanol solvents can cause the polymer surface to swell or partially dissolve to load the catalytic particles on the polymer surface, with strong adhesion. Thereby, silver-iron inks configured with dichloroethane and ethanol as solvents can be directly printed on PET substrates. Zhang et al. [14] reported that selectively metallized photopaper was prepared with catalyst solution printed onto its surface, followed by electroless copper plating. Further, the (NH4)2PdCl4 catalyst solution can bond with the amine groups along the photopaper fiber. These efforts have promoted the continued development of the PE technology. However, we realize that the value of randomness has been neglected in these works.

B. Physical Unclonable Function

PUF was first proposed in 2002 by Pappu et al. [10], which endows physical entities with inherent and unclonable characteristics, enabling it can be used to uniquely identify components. Since the universality of deviations in manufacturing process, various concepts of PUF have been proposed in the past two decades. Electrical PUFs and optical PUFs are the two main branches of PUF, where the former design specific circuits to generate fingerprints, while the latter utilize the rich entropy of images to extract fingerprints [11], [12]. In the PE field, various electrical PUFs have been proposed, including printed memory PUF [15], RO-PUF [2], and PE-PUF [16]. Kuribara et al. [2] proposed a powerful PUF generated from organic ring oscillators (ROs). However, the organic PUF circuit has high complexity, and is prone to yield problem and high operating voltage, which is unsuitable for low-power applications. A PE-PUF based on EGFET is proposed in [16], which is a memory-based circuit that can work at low voltage, and has low circuit complexity. Although the area is reduced compared to RO-PUF, it is still a significant cost for low-cost PE products.

Currently, the latest research on optical PUFs focus on the manufacture of unclonable labels, which can be attached to products and read by optical equipment [17]-[19]. Matsumoto et al. [18] used the special marking techniques to create unique random nanoscale patterns on each object, which truly cannot be duplicated or counterfeited. However, this requires an expensive special etching process, which is not suitable for PE. Benedikt et al. [19] proposed to uses the surface pattern of injection-molded plastic components as PUF to further reduce authentication costs of component. Sharma et al. [20] used image texture instead of the microstructure to authenticate paper, and established a low-cost system. These schemes are completely zero-cost for part serialization, but are limited to specific scenarios, not to mention the PE field. Notably, Erozan et al. [17] proposed an iPUF to conduct counterfeit detection and prevention, which firstly extends the optical PUF to the PE field. Through appearance inspection and image processing of devices in PE system, fingerprints are extracted from the optically visible changes in printing inks. However, the PE devices have the same structure, which makes the optical change and contained information entropy limited, thus the discrimination ability between different devices is insufficient. Moreover, with the proliferation of PE devices, it is hard to guarantee that this method can also effectively distinguish between different devices.

III. The Proposed PE-Based Optical Unclonable Label

In this section, we firstly give a detailed elaboration of the proposed OUL, include the principle of forming unclonable patterns, the selection of printing parameters, and the design of OUL. Then, a robust ID extraction algorithm is presented, which is a key step in determining how well the authentication works, where the printed samples are captured, pre-processed, aligned and located, and convert into unique IDs. Later, a lightweight authentication is explained, and a pseudo label recognition is added. Finally, some evaluation metrics are introduced to assess the performance of OUL.

A. Manufacturing of Optical Unclonable Label

In PE process, PET film has been widely used as substrate due to its good mechanical and optical properties, which is a transparent and glossy film. However, the hygroscopicity and adhesion of its surface is poor. When aqueous ink is printed on PET film, the ink splashes, redistributes and forms droplets of different sizes under the effect of surface tension. This phenomenon makes aqueous nano-silver inks to form complex and unpredictable patterns on PET film. Moreover, the position, size and shape of the ink dots on PET film cannot be manually controlled, which presents good randomness and uniqueness, that is attractive for manufacturing optical PUF.

Among the many printer parameters, drop spacing (DS) has a great influence on the formed pattern, which mainly affects the density of ink droplets, the larger the DS, the sparser the ink, and the smaller the dot size formed. Fig. 1 shows the effect of printed patterns under different DS values, it can be seen that the complexity of pattern increases with the increase of DS value, while the size of each ink dot in the pattern is decreased conversely, that means it requires an advanced camera with higher resolution and sensitivity to photo the pattern. Therefore, one should better tradeoff the DS value between the information entropy and the resolution of the available camera.

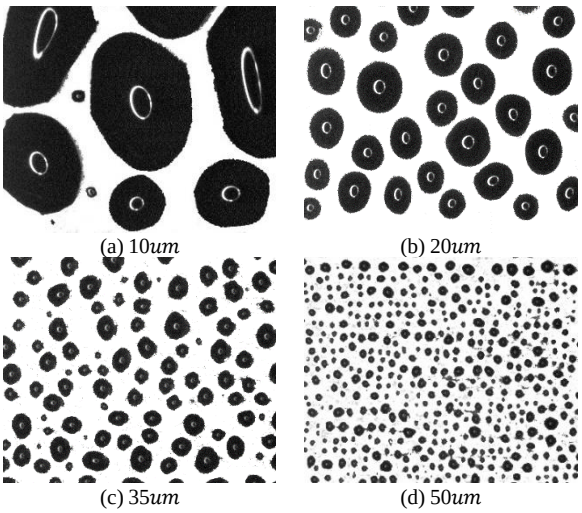


Fig. 1. Patterns of ink droplets printed under different DS values. All the pattern is a $1.5mm * 1.5mm$ square printed under the same conditions.

In our experiment, PET film was used as the substrate, and the aqueous nano-silver ink (nano-silver particle solid content is 5%, particle diameter is 15 nm, and the solvent used is

ethylene glycol and methanol) was used as solution. The material precipitation instrument used for printing is Dimatix DMP-2850 inkjet printer. Fig. 2 shows the designed OUL pattern, which contains two parts: PUF block and location block. PUF block as the main part, is the source of information entropy, which is a square with length of 3mm, with DS value of 20um. Location block as the auxiliary part, is designed for location, which is a rectangle with length of 3mm and width of 0.5mm located 0.5mm above PUF block, with DS value of 5μm.

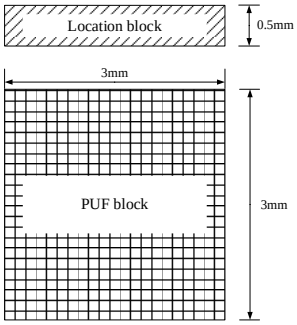


Fig. 2. Schematic of designed OUL.

B. ID Extraction Algorithm

Optical PUFs are often difficult to analyze directly, to quantify the effect of printed OULs, designing a robust ID extraction algorithm is necessary. Fig. 3 depicts the flowchart of proposed ID extraction algorithm. Firstly, a camera is required to photo the printed OUL, obtain the original image. Then, some preprocessing operations are performed to eliminate the influence of environment, such as illumination and camera noise. Next, the PUF block is aligned and the region of interest (ROI) is located, based on the location block. Finally, the response is extracted from the ROI and hashed to a 256-bit sequence. All operations are run in Pycharm 2021.7, and an open-source image processing library OpenCV is used to process the image.

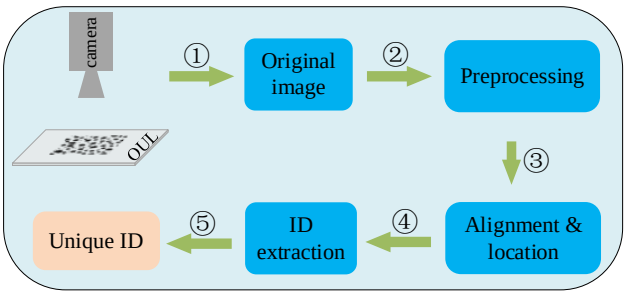


Fig. 3. Flowchart of the ID extraction algorithm.

1) image acquisition: Transmitted light imaging was used to avoid the interference of environmental conditions on the image, that could obtain a clearer image. The commercial industrial camera VILIANRUI S-6 was applied to photo the image, which is a common low-cost ordinary optical camera with a resolution of 2592×1944 pixels. The light source used is a machine vision liquid crystal display (LCD) light source JANPOS JH-FL4040-W.

2) preprocessing: Environment disturbances, such as illumination and camera noise, are inevitably introduced during printing and acquiring the OULs. To increase the reliability of extracted IDs, image grayscale and Gaussian filter were applied, which can weaken the influence of slight change of color and highlight the target zone. Moreover, the images after preprocessing take up less memory that will faster computing speed in the subsequent steps.

3) Alignment and location: The captured original images include the whole OUL and its neighborhood, and will also introduce some rotation, which means some correction and offset are needed. To achieve this purpose, such operations were performed: image binarization, location block detection, transformation matrix generation, and ROI location.

a) Image binarization: Setting the grayscale value of image pixels to 0 or 255, gives the whole image a distinct black and white effect. As the ink dots and the background have distinct difference, they can be easily binarized with an appropriate threshold, which was set to 127 in this algorithm.

b) Location block detection: Contour detection was performed on binarized images, and minimum bounding rectangle (MBR) was adopted to identify the location block. Meanwhile, the vertex coordinates of MBR were recorded to judge whether the image has been rotated or not.

c) Transformation matrix generation: When rotation happens, correction is necessary. The transformation matrix through establishing a relationship between original space and image space to offset the rotation, which contains the mapping rules for rotating to the horizontal direction.

d) ROI location: To obtain the desired ROI, the PUF block was focused and centrally cropped. As the PUF block is located at a fixed position below the location block, which can be directly identified by calculating the coordination.

4) ID generation: Perceptual hash algorithm was used to generate IDs, which endows each image a “fingerprint” string, through comparing them to determine the similarity of the images. Perceptual hash algorithm mainly consists of four steps: scaling the image size, calculating the image threshold, binarizing the grayscale image, and obtaining the hash value.

a) Scaling the image size: Scaling the located ROI to a size of 16×16, a total of 256 pixels, can remove the details while maintaining basic information, such as structure, and attenuate image differences caused by different scales.

b) Calculating the image threshold: To binarize an image, a threshold needs to be selected, and the simplest way is to calculate the average value of the whole image, while ensuring that the pixels are evenly distributed.

c) Binarizing the grayscale image: Comparing the grayscale value of each pixel with the precalculated threshold. If one is greater than the threshold, the corresponding bit of response is recorded as 1; otherwise, it is recorded as 0.

d) Obtaining the hash value: Combining the results of comparison to generate a 256-bit binary sequence, which is used as the fingerprint of this image. In our algorithm, the values are organized in the order of pixel positions.

Fig. 4 illustrates an example of the whole process of ID extraction algorithm, where Fig. 4(a) is the captured image of original OUL. Fig. 4(b) shows the effect after preprocessing, which is visually more focused and the impact of noise is downplayed. Fig. 4(c) shows the principle of location, assuming that the coordinates of the upper left and the lower right corners of the MBR are (a, b) and (m, n) respectively. Then the ROI is a square with side length of *w-h* and its upper left corner locates at $(a + h/2, b + 2h + h/2)$, where *w=m-a*

and $h=n-b$. Fig. 4 (d) shows the location block detection and image alignment, where the OUL was correctly located and aligned. From left to right, Fig. 4(e) illustrates how to convert the located ROI to a 256-bit binary sequence step by step. Notably, ID generation is essentially a hash mapping process that increases reliability while decreasing uniqueness, thus tradeoff between uniqueness and reliability should be further considered.

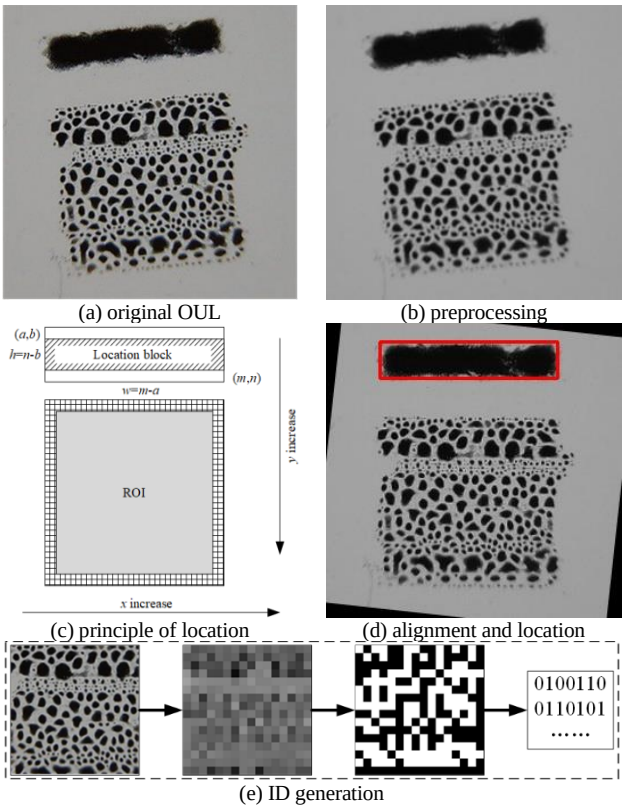


Fig. 4. Example of the whole ID extraction process.

C. Lightweight Authentication System

The proposed OULs are attached to PE products, which are usually mass-produced and resource-constrained, thus the authentication scheme must be lightweight. Fig. 5 depicts the schematic of proposed authentication system, which contains four components: PE products, marked with OULs; the manufacturer, capturing the OULs and extracting unique IDs to establish the database; the end-users, hoping to validate the authenticity of products; and the authentication mechanism, receiving requests and giving feedback. Further, the whole authentication process can be divided into two phases: registration and validation.

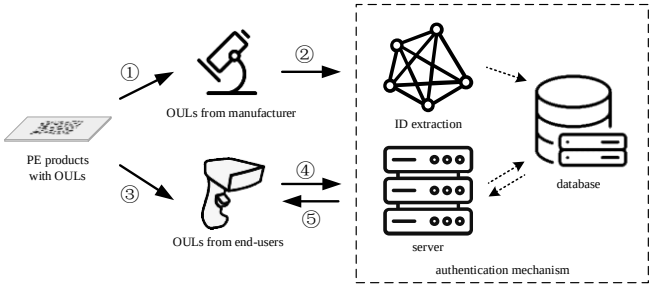


Fig. 5. The proposed lightweight authentication system.

1) Registration phase: During the manufacturing process of PE produces, a special step is added to print the proposed OUL. As shown in Fig. 6(a), once these products are finished, their attached OULs will be captured by the manufacturer, then convert into binary sequences, establishing a products database, which is the basis for subsequent validation.

2) Validation phase. Before validation, the legitimacy of the label reader is checked, and only legitimate readers can communicate with the server. After establishing a trusted link with the authentication server, pseudo label recognition is performed, which utilizes the statistics of the overall sample to resist forged labels. Then the extracted ID is compared with the database, and if the Pearson correlation coefficient is greater than the predefined threshold, it is considered real, otherwise, it is fake.

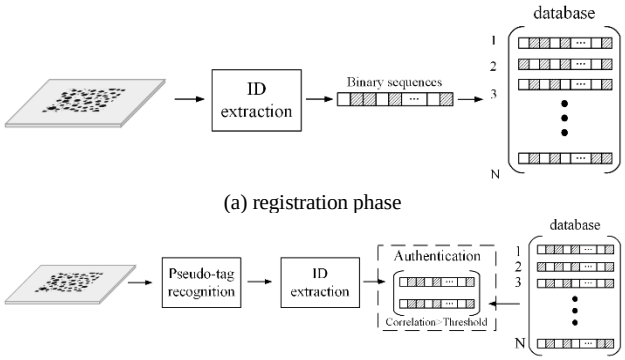


Fig. 6. Two phase of authentication mechanism.

D. Evaluation Metric

To quantitatively analyze the proposed method, some metrics are used to evaluate the designed OUL and ID extraction algorithm, including identification certainty, FoM and overhead. Hamming distance (HD) is commonly used to evaluate the uniqueness (inter HD) and reliability (intra HD), where inter HD represents the Pearson correlation coefficient of two IDs extracted from different samples; intra HD represents the Pearson correlation coefficient of two IDs extracted from the same sample. However, for optical PUF, inter HD and intra HD cannot be fully adapted, thus the identification certainty was used to access the unpredictability of printed OULs. The identification certainty ζ_n calculates the difference between $\mu_{\text{inter},\,n}$ and α_n in the units of $\sigma_{\text{inter},\,n}$, is defined as Eq. (1):

$$\zeta_n = \frac{\alpha_n - \mu_{\text{inter},\,n}}{\sigma_{\text{inter},\,n}} \tag{1}$$

where n is the index number of the samples, μ_{inter} and σ_{inter} are denote the mean value and the standard deviation of inter HD, and α denotes the intra HD of single sample.

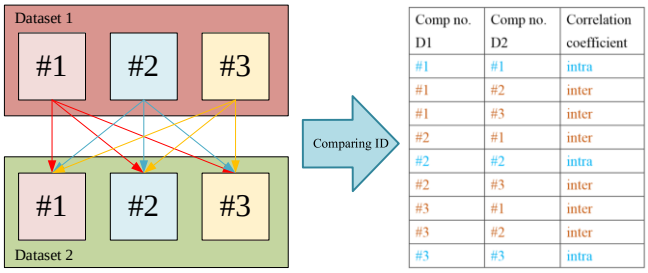
FoM denotes the difference between the minimum value of intra HD and the maximum value of inter HD, which was used to reflect the distinguishability among PUFs [17]. The FoM can be defined as Eq. (2):

$$FoM(I) = \min_{\{(i,j)|i=j,i,j\in I\}} C(i,j) - \max_{\{(i,j)|i\neq j,i,j\in I\}} C(i,j) \tag{2}$$

where I represent the set of all PUF samples, i and j are the index number of the samples; the first summand denotes the minimum value of intra HD, while the second summand denotes the maximum value of inter HD.

IV. Results and Discussion

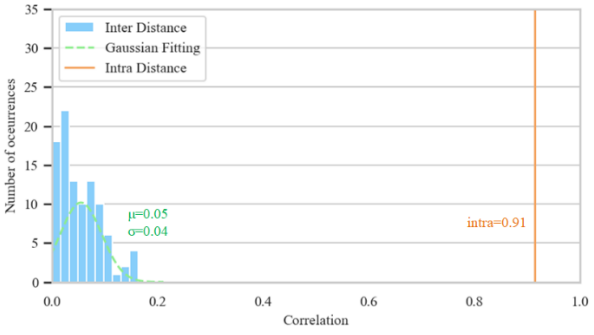
In our experiment, 100 OULs were printed to build a database, evaluating the proposed authentication scheme. Firstly, each OUL was pinned a custom-made sample holder, using the image acquisition device to photo the images, then taken out, re-inserted and repeat. Hence, two sets of OUL images were obtained. Next, processing each image with the proposed ID extraction algorithm to obtain a unique ID. Assume that the IDs extracted from the first OUL images set is dataset1, while the IDs extracted from the second OUL images set is dataset2. After that, calculating the inter HD and intra HD, as illustrated in Fig. 7, comparing each ID in dataset1 with all IDs in dataset2, until all intra HD and inter HD were obtained.



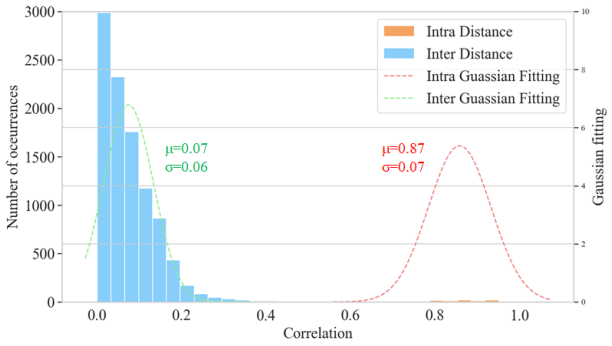
Comp no. D1	Comp no. D2	Correlation coefficient
#1	#1	intra
#1	#2	inter
#1	#3	inter
#2	#1	inter
#2	#2	intra
#2	#3	inter
#3	#1	inter
#3	#2	inter
#3	#3	intra

Fig. 7. Evaluation of the proposed OUL.

1) identification certainty: Fig. 8 shows the distribution of inter HD and intra HD among OULs, where the horizontal axis indicates the correlation; the vertical axis indicates the frequency of occurrence; the blue and orange boxes represent the inter and intra HD, and the green and red dotted line are the corresponding Gaussian fitting curve, respectively. Fig. 8(a) is relative to a single OUL, where the mean value μ_{inter} is 0.05; the standard deviation σ_{inter} is 0.04, and the intra distance α is 0.91. Therefore, the identification certainty of single OUL can be calculated as 21.5σ. Fig. 8(b) is relative to the whole OULs, where the mean value μ_{inter} of 0.07; the standard deviation σ_{inter} of 0.06, and the intra distance α is 0.87. It can be calculated that the identification certainty of the printed samples is 13.3σ. Based on the assumption of Gaussian distribution, the probability of locating outside the interval $\mu+10\sigma$ is less than 10^{-22}, which means that the IDs extracted from the proposed OULs have great randomness and uniqueness.



(a) single OUL



(b) all OULs

Fig. 8. The inter distance and intra distance of all samples.

2) FoM: By controlling the scaling size, we obtained the FoM results under different bit sizes, as shown in Fig. 9, with the increase of bit length, that the FoM initially increases rapidly and then tends to level off. The increase of FoM indicates that the distinguishability between OULs has been improved, however, when the bit length exceeds 256, the FoM rises slowly and begins to fluctuate, meaning that further increases in bit length have little effect. The maximum FoM obtained is 0.29, which is better than the iPUF proposed in [17], with a maximum FoM of 0.15. As compared in Table I, under the same bit length, the FoM of our proposed OUL is much bigger than iPUF, even when the bit length reduced to 256, we can get a FoM of 0.25, which is still better than the maximum FoM that iPUF can obtain, with the bit length of 2400, that means lower storage is needed.

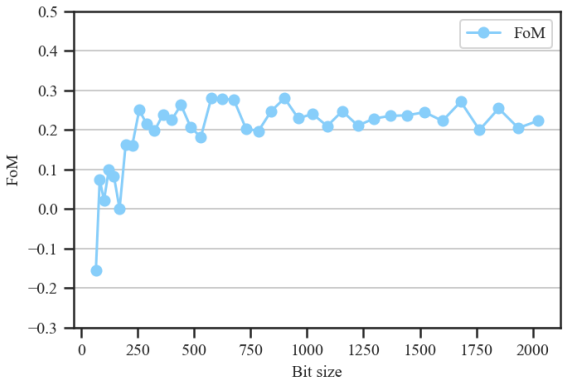


Fig. 9. Distribution of FoM under different bit size.

TABLE I. Max FoM and Required Bit Size of Different Methods

Scheme	Bit size	FoM
iPUF [17]	256	< 0
	2400	0.15
Proposed	256	0.25
	600	0.29

3) overhead: The overhead mainly contains two sources: the additional area overhead of printing OUL and the storage cost of IDs, where the latter is primarily considered in the scenario of a bulk of products. Combining the above analysis of identification certainty and FoM, it can be concluded that different OULs can be effectively distinguished with the bit length of 256. Hence, the storage cost is 256 bits per product, which is significantly reduced compared with original image. As for the area overhead, the size of OUL is $3mm \times 5mm$.

In [17], a camera with microscope component was used to obtain high-resolution, detailed image, which increases the cost and difficulty of identification. While in our scheme, as illustrated in Fig. 1, one can adjust the DS value to increase the information entropy. If a camera with the same accuracy as [17] were used to photo the OUL, the additional area introduced can be reduced to less than $1mm^2$, which will be much easier to print on the blank space of PE substrates.

4) statistics of the overall sample: Despite the ink droplets of the printed OULs exhibit distinct differences in size, contour and position, theoretically, the same batch should present some similar distributions, that can be used for pseudo label recognition. The results confirmed our suspicions, we chose the area of the ink dops, the radius difference between the minimum enclosing circle and the largest inscribed circle of each ink droplet, expressed as $\Delta radius$, and the number of ink droplets, expressed as #ink droplets, in each OUL as the judgment parameters, where the area reflects the size of the ink droplets, and the $\Delta radius$ indicates the irregularity of the ink droplets. Table II summarizes the statistics of the whole printed 100 OULs, including the mean value μ, the standard derivation σ and the corresponding confidence interval. According to the PauTa criterion, in the normal distribution, the probability that the variable value lies in the interval $[\mu - 2\sigma, \mu + 2\sigma]$ is 95.45%, which is usually regarded as the confidence interval, and the area, the $\Delta radius$ and the #ink droplets can be obtained as [0, 334.72], [0, 7.69] and [57.96, 229.48], respectively. When more than 15% of the ink droplets do not satisfy the distribution or the # ink droplets are far away from the confidence interval, regarding the OUL is forged.

TABLE II. DISTRIBUTION OF THREE JUDGMENT PARAMETERS

Parameter	μ	σ	Confidence interval
area	104.60	115.06	[0, 334.72]
$\Delta radius$	2.79	2.45	[0, 7.69]
#ink droplets	143.72	42.88	[57.96, 229.48]

V. CONCLUSION

In this paper, a phenomenon that aqueous nano-silver inks printed on PET film forming complex and unpredictable pattern is discovered and utilized. Initially, we design an OUL, which is compatible with PE technology and only introduces slight area overhead. The pattern of OUL is flexible enough, enabling it release from the constrains of the circuit. Then, a robust ID extraction algorithm is designed to build lightweight authentication system, which can effectively reduce storage cost of OUL without compromising uniqueness and reliability. In addition, a pseudo label recognition is added to resist forged labels, which utilizes the overall sample statistics to provide double insurance. To evaluate the effect of proposed method, 100 OULs are printed, with an overall identification certainty of 14.17σ. Compared with [17], the FoM of our scheme is much improved, with the best of 0.29, which shows great attraction to the current anti-counterfeiting scheme.

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